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BEARING DEVICE WITH INTEGRATED ELECTRIC INSULATION, NOTABLY FOR AN ELECTRIC MOTOR OR MACHINE

Abstract

A bearing device includes a bearing having first and second rings configured to rotate relative to each other and a bushing and an electrically insulating insert. The second ring has first and second cylindrical surfaces. The bushing has first and second cylindrical surfaces. The electrically insulating insert is overmolded between and connects the first cylindrical surface of the bushing and the second cylindrical surface of the second ring. The first cylindrical surface of the bushing includes a first annular band of material projecting toward the second cylindrical surface of the second ring, the first annular band of material has a radial surface facing the second cylindrical surface of the second ring, the first annular band of material has an axial width less than an axial width of the bushing, and the radial surface of the first annular band of material includes at least one planar portion.

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Background/Summary

CROSS-REFERENCE

[0001] This application claims priority to French patent application no. 2402411 filed on Mar. 11, 2024, the contents of which are fully incorporated herein by reference.

TECHNOLOGICAL FIELD

[0002] The present invention relates to the field of bearings that are particularly used in electric motors, electric machines and associated equipment.

BACKGROUND

[0003] In an electric motor or machine, at least one rolling bearing is mounted between the housing of the electric motor or machine and the rotary shaft in order to support this shaft. During operation, when the shaft is rotating, a difference in electric potential can occur between the shaft and the housing of the motor or of the electric machine, generating an electric current between the inner ring of the rolling bearing, which is rigidly connected to the shaft, and the outer ring, which is rigidly connected to the housing. The electric current flowing through the components of the rolling bearing can damage these components, notably the rolling elements and raceways provided on the inner and outer rings. Electric discharges can also generate vibrations.

[0004] In order to overcome these disadvantages, a known solution involves replacing the rolling elements of the bearing, which are made from the same steel as the inner and outer rings, with rolling elements made of ceramic. This is generally referred to as a hybrid rolling bearing. However, such a hybrid rolling bearing is relatively expensive.

[0005] In order to overcome the aforementioned disadvantages, another known solution involves equipping the outer ring of the rolling bearing with an insulating sleeve provided with a bushing and an insulating insert made of electrically insulating material and radially interposed between the outer ring and the bushing. In order to attach the insulating insert to the outer ring and to the bushing without any additional elements or specific machining on the outer ring, the insulating insert can be overmolded. However, with such a solution, relative uncoupling of the insulating insert and of the bushing can occur during operation.

SUMMARY

[0006] Therefore, an aspect of the present disclosure is to overcome the aforementioned disadvantages by providing a bearing device with a simple and economical design.

[0007] The disclosure relates to a bearing device comprising a bearing provided with a first ring and a second ring able to rotate relative to each other.

[0008] The device further comprises at least one insulating sleeve mounted on the second ring of the bearing. The insulating sleeve is provided with a bushing and an insulating insert radially interposed between the second ring of the bearing and the bushing. The insulating insert is made of an electrically insulating material.

[0009] The bushing comprises an outer surface and an inner surface, opposite the outer surface, which delimit the radial thickness of the sleeve. The insulating insert is overmolded onto the second ring of the bearing and at least onto one of the outer and inner surfaces of the bushing.

[0010] According to a general feature, the surface of the bushing comprises at least one protuberance that projects towards the second ring and is provided with a planar portion (a “flat”) radially oriented towards the second ring. This provides a bearing device with integrated electrical insulation that is economical compared with conventional hybrid rolling bearings. Furthermore, the

device is easy to manufacture and assemble into the associated electric motor or machine.

[0011] In addition, providing the protuberance with the planar portion on the bushing allows a good connection to be obtained with the insulating insert insofar as a planar portion with a matching shape is formed on the insert during overmolding. The risk of any relative movements between the insulating insert and the bushing in the circumferential direction is particularly limited, notably during temperature variations. In addition, the protuberance forms an axial stop surface that allows any relative movements between the insulating insert and the bushing in the axial direction to be limited.

[0012] As used herein, “axial direction” is understood to mean the direction parallel to the axis of the bearing device and “circumferential direction” is understood to mean the direction that is perpendicular both to the axial direction and to a radius of the bearing device, in other words, tangent to a circle whose center is on the axis of the bearing device.

[0013] The bushing can be provided with two front faces delimiting the axial length thereof. The protuberance can radially extend one of the front faces. In one embodiment, the planar portion comprises a first circumferential edge that circumferentially connects to an end zone of a cylinder portion of the protuberance. The planar portion can comprise a second opposite circumferential edge that circumferentially connects to another end zone of the cylinder portion of the protuberance, or to an end zone of another cylinder portion of the protuberance.

[0014] According to one design, the protuberance is provided with a single cylinder portion and the planar portion that circumferentially connects to the cylinder portion.

[0015] According to another design, the protuberance is provided with a plurality of planar portions spaced apart from one another in the circumferential direction, and a plurality of cylinder portions that each extend between two successive planar portions.

[0016] According to yet another design, the protuberance is provided with a plurality of planar portions, with at least some or all of the planar portions circumferentially connecting with each other.

[0017] In a particular embodiment, the bushing is made of metal material. The bushing thus can be easily machined to a predetermined radial tolerance.

[0018] In one embodiment, the insulating insert covers the whole of the surface of the bushing. In this case, the insulating insert completely covers the surface of the bushing in the axial direction and in the circumferential direction.

[0019] According to a first design, the bushing delimits the outer surface of the device. In this case, the second ring is the outer ring of the bearing.

[0020] According to a second alternative design, the bushing delimits the inner surface of the device. In this case, the second ring is the inner ring of the bearing.

[0021] If the insulating insert is made of synthetic or elastomer material, it makes the device insensitive to temperature variations.

[0022] In a particular embodiment, the bearing comprises at least one row of rolling elements disposed between raceways of the first and second rings. The rolling elements can be made of metal material.

[0023] The disclosure also relates to an electric motor comprising a housing, a shaft and at least one bearing device as defined above and radially mounted between the housing and the shaft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The present invention will be better understood with reference to the detailed description of an embodiment, which is provided by way of a non-limiting example and is illustrated by the appended drawings, in which:

[0025] FIG. 1 is a half-view of an axial section of a bearing device according to one embodiment of the present disclosure.

[0026] FIG. 2 is a perspective view of a bushing of the bearing device of FIG. 1.

[0027] FIG. 3 is a side view of the bushing of FIG. 2.

[0028] FIG. 4 is a section view along the IV-IV axis of FIG. 3.

DETAILED DESCRIPTION

[0029] The bearing device illustrated in FIG. 1 comprises a bearing 10 provided with a first ring 12 and a second ring 14 that are able to rotate relative to each other about the X-X' axis of the bearing. In the illustrated embodiment, the first ring 12 is the inner ring of the bearing and the second ring 14 is the outer ring. As will be described in further detail hereafter, the bearing device is designed so that it does not conduct electric currents. The bearing device has integrated electric insulation.

[0030] The inner ring 12 and the outer ring 14 of the bearing are concentric and extend axially along the X-X' axis of the bearing and are made of steel. The rings are solid type rings.

[0031] In the illustrated embodiment, the bearing 10 also comprises a row of rolling elements 16, in this case balls, radially interposed between the inner ring 12 and the outer ring 14. The rolling elements 16 are made of steel. The bearing 10 also comprises a cage 17 for maintaining the even circumferential spacing of the rollers 16. The bearing 10 also can be equipped with sealing seals or flanges.

[0032] The inner ring 12 comprises a cylindrical bore 12a, a cylindrical axial outer surface 12b radially opposite the bore, and two opposite radial front faces (not referenced) axially delimiting the bore and the outer surface. The bore 12a and the outer surface 12b delimit the radial thickness of the inner ring 12. The bore 12a forms the inner surface of the inner ring. The inner ring 12 also comprises an inner raceway 18 for the rolling elements 16 that is formed on the outer surface 12b. The raceway 18 is directed radially outwards.

[0033] The outer ring 14 comprises a cylindrical axial outer surface 14a, a cylindrical bore 14b radially opposite the outer surface 14a, and two opposite radial front faces 14c, 14d axially delimiting the bore and the outer surface. The outer surface 14a and the bore 14b delimit the radial thickness of the outer ring 14. The outer ring 14 also comprises an outer raceway 20 for the rolling elements 16 that is formed on the bore 14b. The raceway 20 is directed radially inwards.

[0034] The bearing device also comprises an electrically insulating sleeve 22 mounted on the outer ring 14. The insulating sleeve 22 is mounted on the outer surface 14a of the outer ring 14. The insulating sleeve 22 is rigidly connected to the outer ring 14.

[0035] The insulating sleeve 22 comprises a bushing 24 and an insulating insert 26 radially interposed between the outer ring 14 and the bushing 24. The insulating insert 26 is overmolded onto the outer ring 14 and the bushing 24.

[0036] The bushing 24 is annular and extends axially and in the present embodiment is formed as a single piece. The bushing 24 comprises a cylindrical axial outer surface 24a and a cylindrical bore 24b radially opposite the outer surface 24a, the axis 25 of which is coaxial with the X-X' axis. The bore 24b forms the inner surface of the bushing 24.

[0037] The bushing 24 also comprises two opposite radial front faces 24c, 24d axially delimiting the bore and the outer surface. The front faces 24c, 24d delimit the axial length of the bushing. The outer surface 24a and the bore 24b delimit the radial thickness of the bushing 24. The outer surface 24a of the bushing defines the outer surface of the bearing device 10. In other words, the outer surface 24a defines the outer diameter of the bearing device 10.

[0038] As can be seen in FIGS. 2 to 4, the bore 24b of the bushing comprises a first protuberance 27 configured as a first annular band of material that projects inwardly, i.e., projects towards the outer ring 14 radially from the bore 24b. The protuberance 27 is located at an axial end of the bore 24b of the bushing. The protuberance 27 radially inwardly extends the front face 24c of the bushing. The outer face of the protuberance 27 is coplanar with the front face 24c.

[0039] The bore of the protuberance 27 (the surface facing the second cylinder) has a cylinder

portion **28** with an axis **25**, and a planar portion **30** that circumferentially connects to the cylinder portion **28**. The planar portion **30** is radially oriented towards the outer ring **14**, i.e., radially oriented inwards. In other words, the planar portion **30** is radially located on the side of the outer ring **14**. The planar portion **30** projects from the cylinder portion **28**. The planar portion **30** has a flat shape.

[0040] The planar portion **30** comprises a first circumferential edge **30a** that circumferentially connects to an end zone of the cylinder portion **28**, and a second opposite circumferential edge **30b** in the circumferential direction that circumferentially connects to an opposite end zone of the cylinder portion **28**. The planar portion **30** forms an interruption in the slope relative to the cylinder portion **28** in its connection zones with the cylinder portion. The planar portion **30** in this case extends over the entire axial length of the protuberance **27**.

[0041] In the illustrated embodiment, the planar portion **30** extends in the axial direction.

Alternatively, the planar portion **30** could be inclined relative to the axial direction so as to also fulfil a function of axially retaining the insulating insert. Alternatively, the planar portion **30** could be inclined at two angles assuming a convex shape of the diamond type or assuming a shape whose slopes may or may not alternate.

[0042] In the illustrated embodiment, the bore **24b** of the bushing comprises, at the other axial end thereof, a second protuberance **31** (second annular band of material) that projects inwards, i.e., projects towards the outer ring **14**. The protuberance **31** projects from the bore **24b**. The protuberance **31** extends radially. The protuberance **31** radially inwardly extends the front face **24d** of the bushing. The outer face of the protuberance **31** is coplanar with the front face **24d**.

[0043] In the illustrated embodiment, the protuberance **31** has a smaller radial dimension compared to that of the protuberance **27**. Alternatively, the protuberance **31** could have a radial dimension that is greater than or equal to that of the protuberance **27**. Alternatively, a protuberance **31** may be omitted. In the illustrated embodiment, the protuberance **31** has a cylindrical bore. Alternatively, in a similar way to the protuberance **27**, the protuberance **31** could have at least one planar portion.

[0044] The bushing **24** is advantageously made of metal material. Thus, the outer surface **24a** of the bushing can be easily machined to a predetermined tolerance if required. Preferably, the bushing **24** is made of steel. The bushing **24** can be obtained from a metal sheet by cutting, stamping and rolling. Alternatively, the bushing **24** can be obtained from a tube or from forged/rolled blanks, or even from sintering and stamping.

[0045] The insulating insert **26** is made of electrically insulating material. The insulating insert **26** can be made, for example, of synthetic material, such as PEEK or PA46, or it even can be made of elastomer material, such as rubber, for example.

[0046] The insulating insert **26** is radially interposed between the outer surface **14a** of the outer ring and the bore **24b** of the bushing. The insulating insert **26** covers the outer surface **14a** of the outer ring. In this case, the insulating insert **26** completely covers the outer surface **14a** in the axial and circumferential directions. The insulating insert **26** also covers the bore **24b** of the bushing. The insulating insert **26** in this case also completely covers the bore **24b** in the axial and circumferential directions. The insulating insert **26** covers the cylinder portion **28** and the planar portion **30** of the bore of the protuberance **27**. The insulating insert **26** also covers the bore of the protuberance **31**.

[0047] As previously indicated, the insulating insert **26** is overmolded onto the outer ring **14** of the bearing and onto the bushing **24**. The insulating insert **26** is overmolded onto the outer surface **14a** of the outer ring **14** and onto the bore **24b** and the protuberances **27**, **31** of the bushing **24**.

[0048] The insulating insert **26** is annular. The insulating insert **26** extends axially. The insulating insert **26** comprises an axial outer surface **26a**, a cylindrical bore **26b** radially opposite the outer surface **26a**, and two opposite radial end faces **26c**, **26d** axially delimiting the bore and the outer surface. The front faces **26c**, **26d** axially delimit the insulating insert **26**. The outer surface **26a** and the bore **26b** delimit the radial thickness of the insulating insert **26**. The outer surface **26a** is in radial contact with the bore **24b** of the bushing and the protuberances **27**, **31**. The bore **26b** is in

radial contact with the outer surface **14a** of the outer ring.

[0049] The outer surface **26a** of the insulating insert matches the shape of the bore **24b** of the bushing and the protuberances **27**, **31**, and thus has a stepped shape. In the area of the protuberance **31**, the outer surface **26a** is thus provided with a cylinder portion with a shape that matches the cylinder portion **28** of the bushing, and with a planar portion with a shape that matches the planar portion **30**.

[0050] In the illustrated embodiment, the faces **14c**, **26c**, **24c** and **14d**, **26d**, **24d** of the outer ring, the insulating insert and the bushing are respectively coplanar.

[0051] Alternatively, other arrangements can be provided. For example, the insulating insert **26** could have a limited axial dimension and could remain axially set back from the faces **14c**, **14d** of the outer ring. Alternatively, the insulating insert **26** could have a greater axial dimension and axially project from the faces **14c**, **14d** of the outer ring. In this case, the insulating insert **26** can at least partly cover these faces **14c**, **14d**. As a variant, the insulating insert **26** could at least partly cover the faces **24c**, **24d** of the bushing.

[0052] In another alternative, or in combination, the bushing **24** could axially project from the insulating insert **26** relative to the faces **26c** and **26d**, or could remain axially set back from these faces.

[0053] The bearing device is manufactured as follows.

[0054] In a first step, the bearing **10** and the bushing **24** having the protuberances **27**, **31** are mounted inside a mold that is provided for overmolding the insulating insert **26**. In this position mounted inside the mold, the bushing **24** is radially spaced apart from the outer ring **14** of the bearing. Then, in a second successive step, the insulating insert **26** is overmolded both onto the outer ring **14** of the bearing and onto the bushing **24**. Finally, the bearing device, which is in the form of a unitary assembly, is removed from the mold.

[0055] As previously mentioned, in this embodiment the protuberance **27** of the bore of the bushing comprises a single planar portion **30**. Alternatively, the protuberance **27** could be provided with a plurality of planar portions **30** spaced apart from one another in the circumferential direction, with two successive planar portions being separated by a cylinder portion.

[0056] In another variant, when the protuberance **27** of the bore **24b** of the bushing comprises a plurality of planar portions **30**, provision can be made for several successive planar portions to be circumferentially connected to each other, or even for all the planar portions to be circumferentially connected to each other. In the latter case, the protuberance **27** is devoid of a cylinder portion.

[0057] When the protuberance **27** of the bushing comprises a plurality of planar portions **30**, the planar portions can be identical to each other, or, on the contrary, can have different lengths and/or circumferential dimensions, and/or different inclines.

[0058] In the illustrated embodiment, the first ring **12** of the bearing is the inner ring and the second ring **14**, on which the insulating insert **26** is molded, is the outer ring.

[0059] Alternatively, a reverse arrangement can be provided whereby the second ring **14**, onto which the insulating insert **26** is molded, is the inner ring. In this case, the insulating sleeve is located in the bore **12a** of the inner ring. The insulating insert is then radially interposed between the bore **12a** of the inner ring and the outer surface of the bushing. The insulating insert is overmolded onto the inner ring and at least onto the outer surface of the bushing. The outer surface of the bushing is provided with the one or more flat protuberances. The bore of the bushing delimits the bore of the bearing device.

[0060] In the described embodiments, the bearing of the device is provided with a single row of rolling elements. As a variant, the bearing can be provided with several rows of rolling elements. In addition, the rolling bearing can include types of rolling elements other than balls, for example, rollers. In another variant, the bearing can be a slider bearing devoid of rolling elements.

[0061] Representative, non-limiting examples of the present invention were described above in detail with reference to the attached drawings. This detailed description is merely intended to teach

a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the invention. Furthermore, each of the additional features and teachings disclosed above may be utilized separately or in conjunction with other features and teachings to provide improved insulated bearings.

[0062] Moreover, combinations of features and steps disclosed in the above detailed description may not be necessary to practice the invention in the broadest sense, and are instead taught merely to particularly describe representative examples of the invention. Furthermore, various features of the above-described representative examples, as well as the various independent and dependent claims below, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

[0063] All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

Claims

1. A bearing device comprising: a bearing including a first ring and a second ring configured to rotate relative to each other, the second ring having a first cylindrical surface and a second cylindrical surface radially spaced from the first cylindrical surface, a bushing having an axial length and a first cylindrical surface and a second cylindrical surface radially spaced from the first cylindrical surface of the bushing, and an electrically insulating insert overmolded between and connecting the first cylindrical surface of the bushing and the second cylindrical surface of the second ring, wherein the first cylindrical surface of the bushing includes a first annular band of material projecting toward the second cylindrical surface of the second ring, wherein the first annular band of material has a radial surface facing the second cylindrical surface of the second ring, wherein the first annular band of material has an axial width less than the axial width of the bushing, and wherein the radial surface of the first annular band of material includes at least one planar portion.
2. The bearing device according to claim 1, wherein an axial width of the at least one planar portion is equal to an axial width of the radial surface.
3. The bearing device according to claim 2, wherein the axial width of the at least one planar portion is equal to the axial width of the first annular band of material.
4. The bearing device according to claim 3, wherein an axial face of the first annular band of material is coplanar with an axial face of the bushing.
5. The bearing device according to claim 1, wherein the radial surface of the first annular band of material includes at least one cylindrical portion.
6. The bearing device according to claim 5, wherein the at least one cylindrical portion extends from a first circumferential end of the at least one planar portion to a second circumferential end of the at least one planar portion.
7. The bearing device according to claim 5, wherein the at least one planar portion comprises a first planar portion, and wherein the at least one cylindrical portion extends from a first circumferential end of the first planar portion to a second circumferential end of the first planar portion.
8. The bearing device according to claim 1, wherein the electrically insulating insert comprises an elastomer.
9. The bearing device according to claim 8, wherein the bushing is made of metal.
10. The bearing device according to claim 1, wherein the first cylindrical surface of the bushing

includes a second annular band of material projecting toward the second cylindrical surface of the second ring, and wherein the second annular band of material is axially spaced from the first annular band of material.

11. An electric motor comprising: a housing, a shaft, and at least one bearing device according to claim 1 mounted radially between the housing and the shaft.
